

The Dopamine Receptors Mediating Inhibition of the Sympathetic Vasopressor Outflow in Pithed Rats: Pharmacological Correlation with the D₂-like Type

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(Received 5 April 2011; Accepted 1 July 2011)

Abstract: This study investigated in pithed rats whether dopamine can inhibit the sympathetic vasopressor outflow and analysed the pharmacological profile of the receptors involved. Male Wistar pithed rats were pre-treated with intravenous (i.v.) bolus injections of gallamine (25 mg/kg) and desipramine (50 µg/kg). The vasopressor responses to electrical stimulation of the sympathetic vasopressor outflow (0.03–3 Hz; 50 V and 2 msec.) were analysed before and during i.v. continuous infusions of the agonists dopamine (endogenous ligand), SKF-38393 (D₁-like) or quinpirole (D₂-like). If inhibition was produced by any agonist, then its capability to inhibit the vasopressor responses to i.v. bolus injections of exogenous noradrenaline (0.03–3 µg/kg) was also investigated. Dopamine (3–100 µg/kg min.) inhibited the vasopressor responses to both electrical stimulation and noradrenaline. In contrast, SKF-38393 (10–100 µg/kg min.) failed to inhibit the vasopressor responses to electrical stimulation; whereas quinpirole (0.1–30 µg/kg min.) inhibited the vasopressor responses to electrical stimulation but not those to noradrenaline. The sympatho-inhibition by quinpirole (1 µg/kg min.) remained unaltered after i.v. SCH 23390 (300 and 1000 µg/kg; D₁-like receptor antagonist), but was abolished after i.v. raclopride (1000 µg/kg; D₂-like receptor antagonist). These doses of antagonists did not modify *per se* the sympathetically-induced vasopressor responses. In conclusion, quinpirole-induced inhibition of the sympathetic vasopressor outflow is primarily mediated by activation of dopamine D₂-like receptors.

Dopamine modulates multiple physiological functions by interacting with central and peripheral receptors [1–4]. Considering structural, transductional and pharmacological criteria, dopamine receptors can be classified into D₁-like (comprising the D₁ and D₅ subtypes) and D₂-like (comprising the D₂, D₃ and D₄ subtypes) types [4,5].

Dopamine produces complex cardiovascular effects by interacting with α - and β -adrenoceptors and/or dopamine receptors [4,6–8]. At the vascular level, D₁-like receptors are mainly located on smooth muscle (mediating direct vasodilatation), whereas D₂-like receptors are located on perivascular sympathetic nerves (mediating sympatho-inhibition) [4,9].

Many studies have described the sympatho-inhibitory effects of dopamine on isolated blood vessels and other tissues [3], but no study has reported this effect on the systemic vasculature. For example, when analysing the sympathetic vasopressor outflow in pithed rats: (i) Hietala *et al.* [10] proposed that dopamine receptors may produce sympatho-inhibition, but no selective antagonists were used; and (ii) Fernández *et al.* [11] suggested that some 5-HT₁ receptor antagonists produced sympatho-inhibition via D₂-like receptors. However, these authors did not analyse whether dopamine or selective agonists could produce sympatho-

inhibition in this experimental model. Hence, the present study investigated in pithed rats: (i) whether dopamine, SKF-38393 (D₁-like receptor agonist) or quinpirole (D₂-like receptor agonist) (table 1), can inhibit the vasopressor responses by either stimulation of the sympathetic vasopressor outflow or intravenous (i.v.) bolus injections of exogenous noradrenaline; and (ii) the role of D₁-like or D₂-like receptors by analysing the antagonists SCH 23390 (D₁-like), raclopride and haloperidol (both D₂-like) (table 1) on the above sympatho-inhibition.

Materials and Methods

General methods. All animal protocols were approved by our Institutional Ethics Committee, in accordance with the guide for the Care and Use of Laboratory Animals in the USA. Experiments were carried out in 120 male Wistar normotensive rats (250–280 g). After anaesthesia with diethyl ether and cannulation of the trachea, the rats were pithed by inserting a stainless steel rod as reported earlier [12]. Then, the animals were artificially ventilated with room air using a model 7025 Ugo Basile pump (56 strokes/min.; stroke volume: 20 ml/kg), as previously established [13]. After bilateral vagotomy, catheters were placed in: (i) the left and right femoral veins, for the infusions of agonists and i.v. bolus injections of antagonists, respectively; and (ii) the left carotid artery, connected to a Grass pressure transducer (P23XL), for recording blood pressure. Both heart rate (measured with a 7P4F tachograph) and blood pressure were recorded simultaneously by a model 7D Grass polygraph (Grass Instrument Co., Quincy, MA, USA). Then, the responses to i.v. continuous infusions of vehicle (physiological saline) or the agonists (dopamine, SKF-38393 and quinpirole) were analysed on the

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Table 1.

Binding affinity constants (pK_i) of several agonists and antagonists for dopamine receptors.

	D ₁ -like		D ₂ -like		
	D ₁	D ₅	D ₂	D ₃	D ₄
Dopamine ¹	9	9	~8.2	~8.4	~7.5
SKF-38393 ²	9.0	~9.3	~6.8	~5.3	~6.0
SCH 23390	~9.7	9.5	~6.0	~6.1	~5.5
Quinpirole ²	5.7	–	8.3	~7.6	~7.5
Raclopride	4.7	–	8.7	8.4	5.6
Haloperidol	~7.1	~7.0	8.9	~8.2	8.6

Data taken (as K_i) from Seeman and Van Tol [5]. ~ stands for 'approximately'.

¹the corresponding K_i value for dopamine was determined at the high-affinity state of the D₃ receptor.

²the corresponding K_i values for SKF-38393 and quinpirole were determined at the high-affinity state for all receptors, except for the D₃ receptor.

vasopressor responses to: (i) stimulation of the sympathetic vasopressor outflow (set 1; $n = 90$); or (ii) i.v. bolus injections of exogenous noradrenaline (set 2; $n = 30$). The vasopressor stimulus-response curves (S-R curves) and dose-response curves (D-R curves) elicited by sympathetic stimulation and exogenous noradrenaline were completed in about 30 min., with no change in resting blood pressure. The sympathetic vasopressor stimuli (0.03–3 Hz) and the dosing with noradrenaline (0.03–3 $\mu\text{g/kg}$) were given using a sequential schedule. The body temperature of each pithed rat was maintained at 37°C by a lamp and monitored with a rectal thermometer.

Experimental protocols. Protocol I. Electrical stimulation of the sympathetic vasopressor outflow. In the first set of rats ($n = 90$), the pithing rod was replaced by an electrode enamelled except for 1-cm length, 9 cm from the tip, placing the uncovered segment at T₇–T₉ of the spinal cord to allow selective pre-ganglionic stimulation of the sympathetic vasopressor outflow [14,15]. Prior to electrical stimulation, the animals received gallamine (25 mg/kg, i.v.) to avoid electrically-induced muscular twitching. Ten min. later, the animals were systematically pre-treated with desipramine (50 $\mu\text{g/kg}$, i.v.) before each S-R curve to block noradrenaline reuptake. Under these conditions, the vasopressor responses to lower stimulation frequencies are greater than those elicited without desipramine [16–18]. After 30 min., baseline values of diastolic blood pressure (a more accurate indicator of peripheral vascular resistance) and heart rate were determined. Then, the sympathetic vasopressor outflow was stimulated by applying trains of 10 sec. (monophasic rectangular pulses of 2 msec. and 50 V), at increasing frequencies (0.03, 0.1, 0.3, 1 and 3 Hz). When blood pressure returned to baseline levels, the next frequency was applied. At this point, this set of animals was divided into three groups.

The first group ($n = 44$) was subdivided into eight subgroups that received an i.v. continuous infusion of: (i) physiological saline (three times, 0.02 ml/min.; $n = 6$); (ii) dopamine (3 and 10 $\mu\text{g/kg min.}$; $n = 6$); (iii) dopamine (30 and 100 $\mu\text{g/kg min.}$; $n = 6$); (iv) SKF-38393 (10 and 30 $\mu\text{g/kg min.}$; $n = 4$); (v) SKF-38393 (100 $\mu\text{g/kg min.}$; $n = 4$); (vi) quinpirole (0.1 and 0.3 $\mu\text{g/kg min.}$; $n = 6$); (vii) quinpirole (1 and 3 $\mu\text{g/kg min.}$; $n = 6$); and (viii) quinpirole (10 and 30 $\mu\text{g/kg min.}$; $n = 6$), respectively. Ten min. later, a S-R curve was repeated to analyse their effects on the sympathetically-induced vasopressor responses.

The second group ($n = 22$) was subdivided into five subgroups that received i.v. bolus injections of: (i) 1 ml/kg given twice of saline ($n = 6$) or 1% ascorbic acid (haloperidol vehicle; $n = 4$); or (ii) 300 and 1000 $\mu\text{g/kg}$ of SCH 23390 ($n = 4$), raclopride ($n = 4$) or haloperidol ($n = 4$). Ten min. later, a S-R curve was repeated, to analyse their effects *per se* on the sympathetically-induced responses.

The third group ($n = 24$) was subdivided into five subgroups that received an i.v. bolus injection of: (i) saline (1 ml/kg; $n = 6$); (ii) SCH 23390 (300 $\mu\text{g/kg}$; $n = 4$); (iii) SCH 23390 (1000 $\mu\text{g/kg}$; $n = 4$); (iv) raclopride (300 $\mu\text{g/kg}$; $n = 6$); and (v) raclopride (1000 $\mu\text{g/kg}$; $n = 4$), respectively. After 10 min., all subgroups received an i.v. infusion of quinpirole (1 $\mu\text{g/kg min.}$) and 10 min. later, a S-R curve was repeated.

Protocol II. Administration of exogenous noradrenaline. The second set of rats ($n = 30$) was prepared as described earlier (including pre-treatment with 50 $\mu\text{g/kg}$ desipramine before each D-R curve), but the pithing rod was left throughout the experiment. Once baseline diastolic blood pressure and heart rate were determined, vasopressor responses were elicited by giving i.v. bolus injections of exogenous noradrenaline (0.03, 0.1, 0.3, 1 and 3 $\mu\text{g/kg}$). Subsequently, this set of animals was divided into six groups receiving i.v. continuous infusions of: (i) saline (three times; 0.02 ml/min.; $n = 6$); (ii) dopamine (3 and 10 $\mu\text{g/kg min.}$; $n = 6$); (iii) dopamine (30 and 100 $\mu\text{g/kg min.}$; $n = 6$); (iv) quinpirole (0.1 and 0.3 $\mu\text{g/kg min.}$; $n = 4$); (v) quinpirole (1 and 3 $\mu\text{g/kg min.}$; $n = 4$) and (vi) quinpirole (10 and 30 $\mu\text{g/kg min.}$; $n = 4$), respectively. Ten min. later, a D-R curve was repeated during the infusion of these compounds.

Other procedures applying to protocols I and II. The doses of dopamine, SKF-38393 and quinpirole were infused at a rate of 0.02 ml/min. by a WPI model sp100i pump (World Precision Instruments Inc., Sarasota, FL, USA). The interval between the different stimulation frequencies/doses of noradrenaline was dependent on the duration of the resulting vasopressor responses (5–15 min.), as we waited until diastolic blood pressure had returned to baseline values.

Compounds. Apart from the anaesthetic (diethyl ether), the compounds used in this study were as follows: desipramine hydrochloride, gallamine triethiodide, noradrenaline hydrochloride, dopamine hydrochloride, quinpirole dihydrochloride, raclopride tartrate and ($\pm$)-1-phenyl-2,3,4,5-tetrahydro-(1H)-3-benzazepine-7,8-diol hydrochloride (SKF-38393) (Sigma Chemical Co., St. Louis, MO, USA); haloperidol (gift: Janssen-Cilag, Mexico City, Mexico) and (*R*)-(+)-7-chloro-8-hydroxy-3-methyl-1-phenyl-2,3,4,5-tetrahydro-1H-3-benzazepine hydrochloride, (SCH 23390) (TOCRIS, Avomouth, Bristol, UK).

All compounds were dissolved in saline, except haloperidol (dissolved in 1% ascorbic acid in saline); these vehicles had no effect on baseline diastolic blood pressure and heart rate (not shown). The doses mentioned in the text refer to the salts of substances, except in the case of all agonists and haloperidol (referred to as free base).

Data presentation and statistical evaluation. All data are presented as means $\pm$ S.E.M. The vasopressor responses to either sympathetic stimulation or exogenous noradrenaline in saline- and agonist-infused animals were determined. The difference between the vasopressor responses within one subgroup of animals was evaluated with Student-Newman-Keul's test, once a two-way repeated measures analysis of variance revealed that the samples represented different populations [19]. Statistical significance was accepted at $p < 0.05$ (two-tailed).

Results

Systemic haemodynamic variables.

Baseline diastolic blood pressure and heart rate were 62 ± 5 mmHg and 329 ± 10 beats/min. ($n = 120$). After the first desipramine injection, both variables increased, but returned to baseline after 5 min. (i.e. 60 ± 3 mmHg and 330 ± 5 beats/min.); these values remained unchanged during the infusions of saline, SKF-38393 or quinpirole (not shown).

Similar results were obtained after the i.v. bolus of saline, 1% ascorbic acid, SCH 23390, raclopride and haloperidol or subsequent desipramine treatment (not shown). In contrast, dopamine infusions (3 and 10 $\mu\text{g/kg min.}$; 30 and 100 $\mu\text{g/kg min.}$) produced sustained tachycardic responses (20 ± 10 and 30 ± 10 ; 50 ± 10 and 60 ± 10 beats/min., respectively) without affecting diastolic blood pressure (not shown).

Initial effects by sympathetic stimulation or exogenous noradrenaline

The onset of the vasopressor responses to sympathetic stimulation or noradrenaline was frequency- or dose-dependent (fig. 1) and immediate. Moreover, noradrenaline produced dose-dependent tachycardic responses (not shown), as previously shown [17]. In all cases, the vasopressor responses to either electrical stimulation or noradrenaline were significant

when compared with their baseline values (not shown). The electrically-induced vasopressor responses are because of selective sympathetic stimulation, as only negligible changes in heart rate were observed, as shown earlier [15,20,21].

Effect of continuous infusions of saline or the dopamine receptor agonists on the vasopressor responses to electrical stimulation or exogenous noradrenaline

Figure 1 shows the vasopressor responses to either sympathetic stimulation (S–R curves; upper panels) or noradrenaline (D–R curves; lower panels) before (control) and during i.v. infusions of saline (0.02 ml/min.) or dopamine (3–100 $\mu\text{g/kg min.}$). In animals infused with saline, both the sympathetically-induced (fig. 1A) and noradrenaline-induced (fig. 1D) vasopressor responses remained unchanged when a subsequent S–R or D–R curve was repeated. In contrast, the infusions of dopamine produced a significant inhibition of the sympathetically-induced (fig. 1B,C) and noradrenaline-induced (fig. 1E,F) vasopressor responses; this inhibition was more pronounced at higher frequencies of stimulation (0.3–3 Hz) and at most doses of noradrenaline (0.1–3 $\mu\text{g/kg}$).

Moreover, the sympathetically-induced vasopressor responses remained unaltered during infusions of saline (fig. 2A) or SKF-38393 (fig. 2B,C). That is why the responses to noradrenaline were not analysed during infusions of SKF-38393.

Unlike saline (fig. 3A), 0.1–3 $\mu\text{g/kg min.}$ quinpirole produced a dose-dependent sympatho-inhibition (fig. 3B,C), while 10–30 $\mu\text{g/kg min.}$ produced no further inhibition (fig. 3D). In contrast, all quinpirole doses failed to inhibit the vasopressor responses to noradrenaline (fig. 3F–H). As 1 $\mu\text{g/kg min.}$ quinpirole produced a marked sympatho-inhibition (fig. 3C), this dose was chosen for further pharmacological analysis.

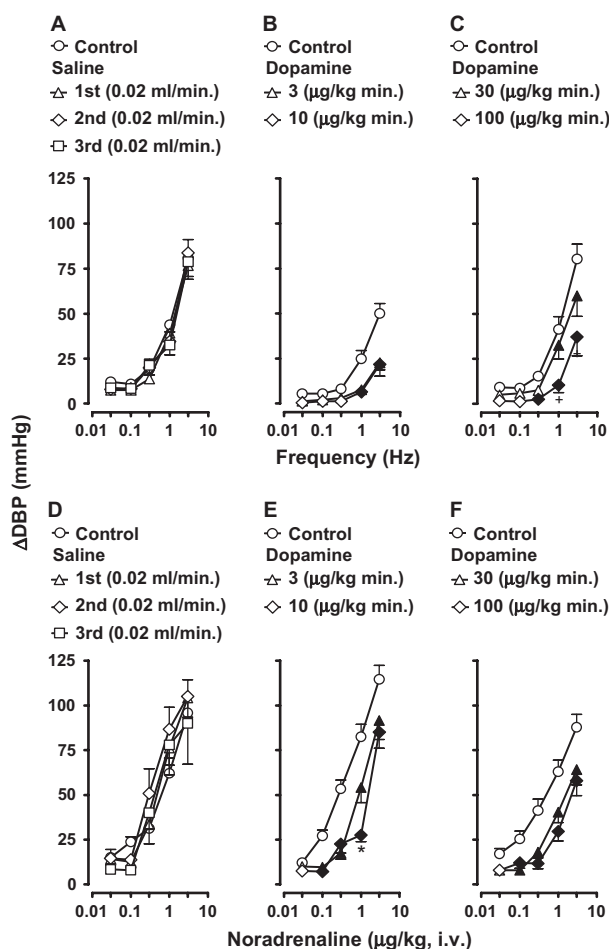


Fig. 1. Vasopressor responses to electrical stimulation (A, B and C) or intravenous (i.v.) bolus injections of exogenous noradrenaline (D, E and F) elicited during continuous infusions of vehicle (saline; 0.02 ml/min.) or dopamine (3 and 10 $\mu\text{g/kg min.}$; 30 and 100 $\mu\text{g/kg min.}$) ($n = 6$ each group). Empty symbols depict either control responses (○) or non-significant responses (△◇□) versus control. Solid symbols (▲◆) represent significantly different responses ($p < 0.05$) versus control. * $p < 0.05$ versus dopamine (3 $\mu\text{g/kg min.}$); + $p < 0.05$ versus dopamine (30 $\mu\text{g/kg min.}$). Δ DBP stands for 'increase in diastolic blood pressure'.

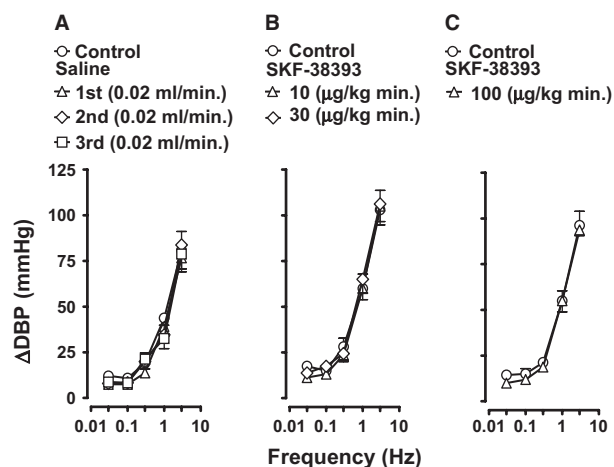


Fig. 2. Vasopressor responses induced by electrical stimulation during continuous infusions of saline [(A) 0.02 ml/min.] or SKF-38393 [(B) 10 and 30 $\mu\text{g/kg min.}$; or (C) 100 $\mu\text{g/kg min.}$]. The control group with saline (A) is the same as that depicted in fig. 1A, but it is shown for the sake of clarity when making comparisons. Δ DBP stands for 'increase in diastolic blood pressure'.

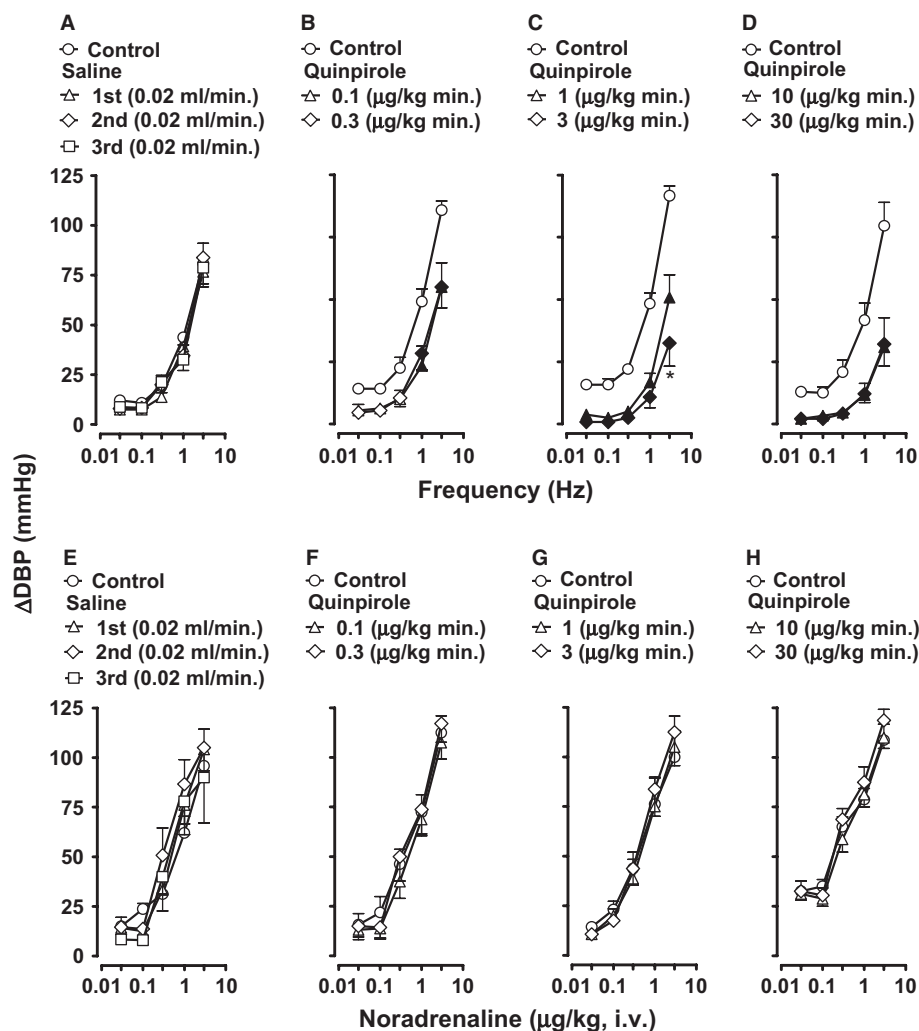


Fig. 3. Vasopressor responses to electrical stimulation (upper panel) or intravenous (i.v.) bolus injections of exogenous noradrenaline (lower panel) elicited during continuous infusions of vehicle (A and E: saline; 0.02 ml/min.) or quinpirole (B and F: 0.1 and 0.3 $\mu\text{g/kg min.}$; C and G: 1 and 3 $\mu\text{g/kg min.}$; and D and H: 10 and 30 $\mu\text{g/kg min.}$). Empty symbols depict either control responses ($\circ$) or non-significant responses ($\triangle$, $\diamond$, $\square$) versus control. Solid symbols ($\blacktriangle$, $\blacklozenge$) represent significantly different responses ($p < 0.05$) versus control. * $p < 0.05$ versus quinpirole (1 $\mu\text{g/kg min.}$). The control group with saline (fig. 3A) is the same as those depicted in figs 1A and 2A, but it is shown for the sake of clarity when making comparisons (likewise for figs 3E and 1D). Δ DBP stands for 'increase in diastolic blood pressure'.

Effect of vehicles or antagonists on the sympathetically-induced vasopressor responses per se or on the sympatho-inhibition by quinpirole.

Figure 4 shows that the sympathetically-induced vasopressor responses remained unchanged after 1 ml/kg of saline (fig. 4A) or 1% ascorbic acid (fig. 4B), as well as after 300 and 1000 $\mu\text{g/kg}$ of SCH 23390 (fig. 4C) or raclopride (fig. 4D). In contrast, 300 and 1000 $\mu\text{g/kg}$ haloperidol blocked these responses (fig. 4E); so it was not analysed on the quinpirole-induced sympatho-inhibition.

Furthermore, fig. 5 shows the effects of i.v. saline, SCH 23390 or raclopride on the sympatho-inhibition by quinpirole (1 $\mu\text{g/kg min.}$). Hence, quinpirole-induced sympatho-inhibition, remaining unchanged after saline (fig. 5A), was as follows: (i) unaffected after 300 and 1000 $\mu\text{g/kg}$ SCH 23390 (fig. 5B,C); (ii) partially blocked after 300 $\mu\text{g/kg}$

raclopride (fig. 5D); and (iii) abolished after 1000 $\mu\text{g/kg}$ raclopride (fig. 5E).

Discussion

General.

Apart from the implications discussed later, our study shows that D₂-like receptors inhibit the sympathetic vasopressor outflow. Interestingly, haloperidol (a less selective D₂-like receptor antagonist; table 1) attenuated *per se* the sympathetically-induced vasopressor responses (fig. 4E), which are mainly mediated by α_1 -adrenoceptors [18]. Hence, haloperidol is not appropriate for analysing the pharmacological profile of receptors mediating sympatho-inhibition in our study, as similarly shown for other antagonists [18]. This attenuation by haloperidol could be due to blockade of vascular

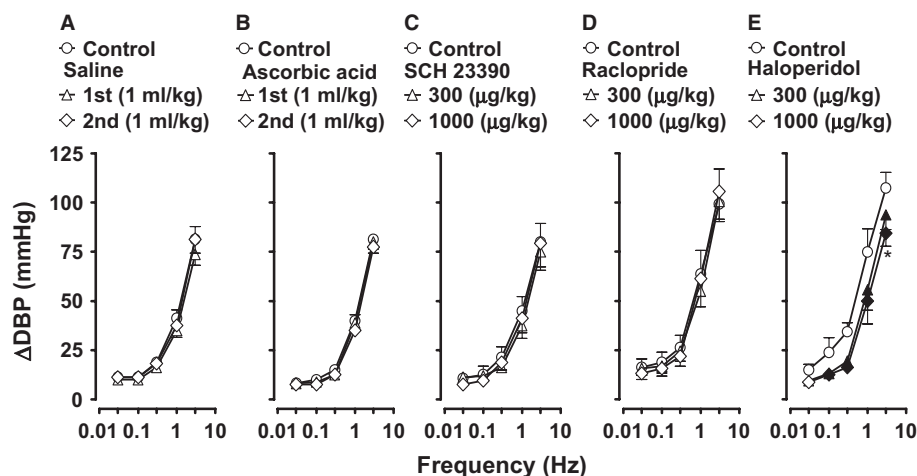


Fig. 4. Effect of intravenous (i.v.) bolus injections of saline (A: 1 ml/kg), 1% ascorbic acid (B: 1 ml/kg), SCH 23390 (C: 300 and 1000 $\mu\text{g/kg}$), raclopride (D: 300 and 1000 $\mu\text{g/kg}$) and haloperidol (E: 300 and 1000 $\mu\text{g/kg}$) *per se* on the vasopressor responses induced by electrical stimulation. Empty symbols depict either control responses or non-significant responses *versus* control. Solid symbols represent significantly different responses ($p < 0.05$) *versus* control. * $p < 0.05$ *versus* haloperidol (300 $\mu\text{g/kg}$). Δ DBP stands for 'increase in diastolic blood pressure'.

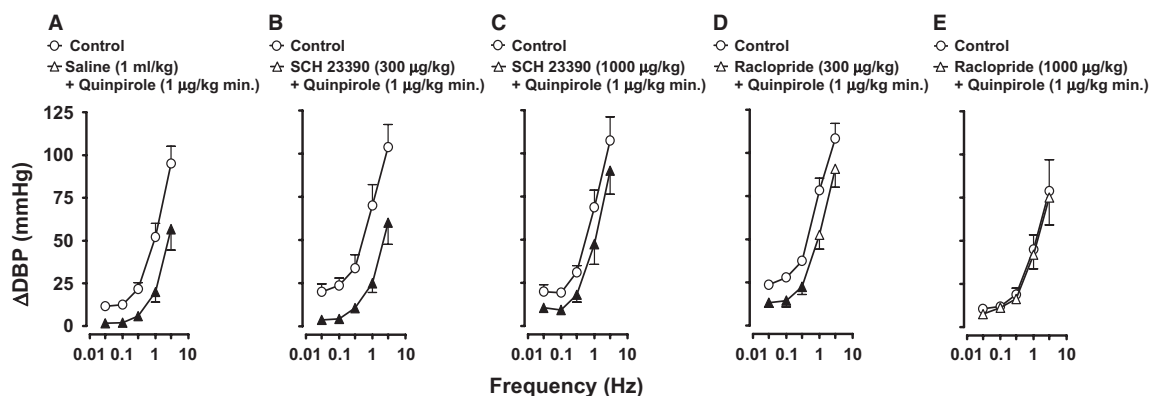


Fig. 5. Effect of intravenous (i.v.) bolus injections of saline (A: 1 ml/kg), SCH 23390 (B and C: 300 and 1000 $\mu\text{g/kg}$, respectively) and raclopride (D and E: 300 and 1000 $\mu\text{g/kg}$, respectively) on the inhibition by quinpirole (1 $\mu\text{g/kg min.}$) of the electrically-induced vasopressor responses. Empty symbols depict either control responses (○) or non-significant responses (△) *versus* control. Solid symbols (▲) represent significantly different responses ($p < 0.05$) *versus* control. Δ DBP stands for 'increase in diastolic blood pressure'.

α_1 -adrenoceptors, for which it displays a reasonable affinity ($pK_i = 8.1$) [22].

Admittedly, sympathetic nerve activity (i.e. the electrically-induced neurotransmitter release) in our study was estimated indirectly by measurement of the evoked vasopressor response [15]. So, the responses to quinpirole were considered sympatho-inhibitory as it inhibited the vasopressor responses to sympathetic stimulation, but not those to i.v. noradrenaline (fig. 3). In contrast, dopamine: (i) inhibited the vasopressor responses to both sympathetic stimulation and noradrenaline (fig. 1); and (ii) can stimulate α - and β -adrenoceptors [7]. Thus, the inhibitory actions of dopamine, regardless of their *loci*, cannot be exclusively because of stimulation of dopamine receptors.

Systemic haemodynamic changes.

That blood pressure and heart rate were transiently increased after desipramine may be attributed to an inhibition of noradrenaline reuptake [23]. Moreover, the potentia-

tion of sympathetically-induced vasopressor responses after desipramine made it possible to reveal more marked sympatho-inhibitory responses to quinpirole at lower stimulation frequencies (fig. 3), as previously shown for other agonists [15,17,18]. This inhibition by quinpirole is not because of tachyphylaxis, as the sympathetic responses remained unchanged in animals receiving saline (fig. 1A). Thus, no time-dependent changes occurred in the haemodynamic variables during the course of our experimental protocols.

On the other hand, the fact that SCH 23390 or raclopride had no effects on baseline blood pressure and heart rate (not shown) or on the sympathetically-induced vasopressor responses (fig. 4) suggests that: (i) these compounds, at the doses used, are devoid of any effect *per se* on the sympathetic responses; and (ii) any effect of a given antagonist on quinpirole-induced sympatho-inhibition is because of a direct interaction of the antagonist with its receptor on the sympathetic nerves.

Considerations on the use of dopamine to inhibit the sympathetic vasopressor outflow.

As dopamine inhibited the vasopressor responses to both sympathetic stimulation and noradrenaline (fig. 1), it is unsuitable to identify the receptors that inhibit the sympathetic vasopressor outflow. Given that it may also interact with peripheral α - and β -adrenoceptors [7], the inhibition by dopamine on the sympathetic responses (fig. 1B,C) may involve stimulation of sympatho-inhibitory D₂-like receptors and/or α_2 -adrenoceptors. Clearly, additional experiments performed with raclopride, rauwolscine and the combination of both antagonists will be required to explore this possibility.

The inhibition by dopamine on noradrenaline-induced vasopressor responses (fig. 1E,F) may involve a physiological antagonism because of vasodilatation via stimulation of vascular β_2 -adrenoceptors [7]. This did not allow us to analyse the receptors mediating the inhibition by dopamine of the sympathetically-induced vasopressor responses.

The role of sympatho-inhibitory D₂-like receptors.

The pharmacological profile of the dopamine receptors that inhibit the sympathetic vasopressor outflow in our study closely resembles that of the D₂-like type in view of: (i) the capability of raclopride to abolish (at 1000 μ g/kg) the sympatho-inhibition to quinpirole (fig. 5E); (ii) the failure of SCH 23390 to abolish (at 1000 μ g/kg) quinpirole-induced sympatho-inhibition (fig. 5C) at doses that completely block D₁-like receptors [24]; and (iii) the fact that SKF-38393 failed to inhibit the sympathetic responses, even at very high doses (fig. 2C). Furthermore, based on the highest affinities of raclopride for the D₂ and D₃ receptor subtypes (table 1), either (or both) subtypes are likely to mediate the vascular sympatho-inhibition in our study.

Possible locus of the sympatho-inhibitory D₂-like receptors.

All D₂-like receptors are coupled to G_{i/o} proteins [4,6] and this is a signal transduction system generally associated with sympatho-inhibition [16,25–27]. Although we have no direct evidence that this is operative in pithed rats, one might speculate upon the possible locus of the sympatho-inhibitory D₂-like receptors in our study. Central mechanisms are excluded as pithed rats were used, but we cannot rule out an action of quinpirole at both the sympathetic ganglia and post-ganglionic sympathetic neurons, which have modulatory D₂-like receptors [1,4,28]. Admittedly, further studies will be required to ascertain the functional role of the D₂-like receptor subtypes (i.e. D₂, D₃ and D₄) on autonomic ganglia and post-ganglionic sympathetic neurons by using subtype-selective antagonists.

In conclusion, our results suggest that the D₂-like (rather than the D₁-like) receptor type mediates inhibition of the sympathetic vasopressor outflow in pithed rats.

Acknowledgements

We thank Dr. Araceli Sánchez-López for her skilful technical assistance and CONACyT (Mexico City) for their financial support (grant 60789-M).

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